Personalized Educational Assistant for Students with Learning Disabilities

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Abstract. Students with learning disabilities often tend to struggle with attention and information retention during regular lessons. Although note-taking and attention tracking would bridge these gaps, actual learning necessitates continuous assistance based on individual needs. Through the implementation of real-time content processing, de- tection of distractions, this research seeks to facilitate easier and more individualized learning by keeping them attentive and noting key points. It introduces the development of a Virtual Educational Assistant for Students with learning disabilities, such as dyslexia, aimed at improving learning opportunities for them through real-time feed- back on attention and understanding. The system incorporates two major functionalities Distraction Detection and Speech Processing. The Distraction Detection module uses computer vision techniques, such as face detection and Eye Aspect Ratio (EAR) calculation, to monitor the attention levels of the students. If any distraction is detected, a voice based alert is given to gain the focus of the student. The Speech Processing module transcribes audio content in real-time or audio file using automatic speech recognition and then summarizes the text for easier understanding. The integration of both modules offers a comprehensive tool that can significantly improve the learning experience for students with learning disabilities by ensuring they remain attentive and improve in academics.With multithread- ing the system output was returned in 25 seconds, and without multithreading it consumed a lot of processing time. Multithreading was used to ensure that the communication between both modules has been in sync for a smooth and faster execution.

Keywords: Learning Disabilities, Personalized Education, Operating Systems, Real-Time Processing, Distraction detection, Speech Processing

1 Introduction

Learning disabilities like Dyslexia [Zingoni et al., 2024], affects some portion of the student population globally, leading to various challenges in academic environments. Students with learning

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disabilities often struggle to stay focused during lectures, understand complex concepts, and understand the information presented. In traditional educational environments, where attention and focus were crucial for learning, these students faced greater difficulties in keeping up with the pace of instructions. As a result, their academic performance tends to decline, leading to a sense of dis- sapointment and frustration. The need for adaptive technologies that can support these students in real-time has become more relevant, as existing methods may not provide that much flexibility or support required to meet their needs.

The recognition of the fact that traditional teaching [Zhang and Zhang, 2024] methods were not al- ways effective for students with learning disabilities, especially those who struggle with attention difficulty, motivated towards this respective research. In the common classroom, these students could find it challenging to maintain focus during long lectures, and without proper support, they may miss critical information. Additionally, the lack of real-time assistance can prevent these stu- dents from actively engaging with the material. Most existing tools for learning disabilities focus on post-session support, such as recorded lectures or textbook-based interventions, but they do not offer the level of real-time interaction needed to prevent students from losing focus in the first place.

Addressing this issue required a system that not only helps students stay engaged or attentive during lessons but also needed immediate support when distractions are detected. This study is motivated by the belief that a combination of real-time distraction detection and speech processing can provide an adaptive solution for students with attention difficulties. By integrating these tech- nologies, the proposed system will allow for a continuous monitoring of students attention, making the learning experience more effective and personalized [Sajja et al., 2024a].

The conventional methods such as taped lectures and extra tutoring [Gutjahr et al., 2017], offer less help, mostly after the lecture has occurred. Such resources fail to provide real-time, immediate support needed to help students maintain focus in classrooms. Such methods also fail to meet individual needs during learning, which is vital in students with attention difficulty. There exists a gap in solutions on the market that this study aims to fill by offering a system to monitor and improve student interest during lessons in real-time.

This study suggests a virtual learning [Essel et al., 2022] aid designed to assist students with learn- ing disabilities by tracking their level of attention and giving instant feedback when distraction is sensed. The system is composed of two main modules: the Distraction Detection module and the Speech Processing module. The Distraction Detection module monitors a student’s involve- ment based on behavioral data, while the Speech Processing module converts class discussions into text,gives live summaries and important points. These are designed to collaborate and keep the students interested in the lesson and that they can be able to retrieve and

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access relevant infor- mation.

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The key objectives of the proposed system are to develop a distraction detection system that mon- itors a student's attention during lessons, provide real-time transcription of classroomdiscussions, offer summaries and key takeaways from the lesson. These objectives aim to create a more effec- tive learning environment by providing personalized support [Chiu et al., 2024] for students with learning disabilities in real-time.

The primary contribution of this study is the combination of distraction detection and speech processing technologies to offer real-time assistance to students with learning disabilities [D'Urso and Sciarrone, 2024]. As compared to the traditional tools that hardly offer any post-session intervention, this system provides constant guidance, allowing students to remain engaged and alert throughout the lesson. With a mix of attention-tracking and speech-to-text technology, this publica- tion adds to more flexible and individualized learning for students with difficulties with attention. The paper is structured as follows: Section II provides a review of related work in educational tools for students with learning disabilities and attention detection methods. Section III details the identified gaps and Section IV tells the methodology of the proposed system. Section V presents the results of the system's implementation, followed by a discussion of its effectiveness. Finally, Section VI concludes the paper and suggests potential directions for future work.

2 Literature Review

2.1 Leveraging AI Technologies for Personalized Learning Support in Dyslexic Students

This work [Meenakshi et al., 2024a] takes a thoughtful look at how artificial intelligence can be used to support students with dyslexia in more meaningful ways. It points out that traditional teach- ing methods often don't work well for these learners, who may struggle with reading, writing, and comprehension. While assistive technologies exist, they're rarely tailored to individual needs. To change that, the authors propose a smart learning system powered by AI using tools like natural language processing, machine learning, speech recognition, and text to speech to create a person- alized and adaptive learning experience. The system is designed to respond in real time to how a student is doing, offering content and support at just the right pace. Behind the scenes, it uses important operating system concepts like multitasking to handle simultaneous activities, resource management to keep things running smoothly, and real-time processing to ensure quick, helpful feedback. The work shows how combining smart software with OS efficiency can make learning more accessible and a lot more empowering for dyslexic students.

2.2 AI-Powered Personalized Learning Assistant

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The work [Lohar et al., 2024] examines a novel learning tool that was designated as the Personalized AI Learning Assistant, with the purpose to redefine the way students interact with learning

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materials. By employing the techniques such as machine learning, natural Language processing and adaptive learning algorithms, offers a highly customized learning experience. Developed using tools like React, Node.js, Flask, and TensorFlow.js Platform tailors content delivery according to the unique preferences of each individual student, and learning style. It does not just deliver con- tent—and also interactive exercises, in real-time feedback, and per Performance metrics to keep learners on track and engaged. It is a computer-based system that is acces possible across devices, prioritizing ease of use, data security and scalability. Overall, the work emphasizes how this AI as- sistant fills the gap between impersonal education and individualized support, delivering students a smarter, more responsive path to attaining their academic aspirations.

2.3 An Adaptive E-learning Platform For Primary School Children With Visual Impairment

This work [Meenakshi et al., 2024b] deals with the problem statement of en Ensuring available e- learning platforms to visually impaired students, highlighting the importance of targeted solutions to overcoming their special learning challenges. It addresses a number of methods and approaches discussed in earlier search to enrich the online learning experience, including the incorporation of learning technology and new teaching techniques, along with the emergence of voice-based tools. While the review does not explicitly specify Operating System (OS) concepts, it touches on the relevance of OS Accessibility features and voice-based interaction capabilities as key factors in the design and application of inclusive e-learning systems. The review highlights the significance of continued research to gauge the influence of these technologies along the learning path of visually impaired students.

2.4 Artificial Intelligence-Enabled Intelligent Assistant for Personalized and Adaptive Learning in Higher Education

This work [Sajja et al., 2024b] addresses the demand for personalized and adaptive learning in higher education by developing an AI-enabled intelligent assistant to improve the learning experience. The focus is on reducing cognitive load and providing tailored support to students. The study employs advanced AI and natural language processing (NLP) techniques to create an interactive learning platform, which is evaluated for its effectiveness. Operating system concepts are applied in several ways: concurrency manages multiple real-time data processing tasks, memory management ensures efficient handling of large datasets, security and privacy measures protect sensitive student data, and integration with system APIs enables seamless communication between the assistant and educational platforms, collectively enhancing the system's

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functionality and user experience.

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2.5 Identified Gaps

Assistive learning tools have evolved a lot, yet there is still a perceivable shortfall when it comes to effectively harnessing the potential of operating system (OS) [Doney, 2025] principles. Concepts such as resource management, scheduling, and prioritization, that are pivotal in how contemporary Operating Systems operate efficiently are not applied to facilitate students with learning disabil- ities. Suppose that these tools were able to handle cognitive load like an OS manages system resources, it would make learning so much more accessible and tailored. And yet, while AI is in- creasingly being utilized in education, it's generally for the average student population. There's so much potential untapped in leveraging AI to customize content, pace, and assistance just for learn- ers with special challenges. Combining these OS principles with careful, AI-based design might render assistive tools more responsive and inclusive.
This study hopes to fill these gaps [Laak and Aru, 2025] by creating a Personalized Educational Assistant that combines OS-level optimizations.

3 Methodology

This section outlines the methodology used to develop the Educational Assistant for Students with Learning Disabilities. The methodology is divided into two key components: Distraction Detection [Mohammed et al., 2024] and Speech Processing.Fig. 1 is the workflow of these two modules.

3.1 Distraction Detection

The Distraction Detection [Asish et al., 2022] module monitors the student's attention level dur- ing educational activities or lectures using real-time computer vision techniques. This module is essential for providing feedback on the student's focus level during tasks.

Face Detection and Landmark Tracking For monitoring the student's attention, the system captures video using a webcam feeds. First, the biggest face in the frame is recognized, then Me- diaPipe's face mesh model [Jakhete and Kulkarni, 2024] is utilized to detect and follow facial movements, with the eyes in focus to check the student's attention state.

Eye Aspect Ratio (EAR) Calculation The system continues to calculate the Eye Aspect Ratio (EAR) [Venugopal et al., 2016], which helps in identifying whether the eyes of the student are closed or if they blink excessively. If the EAR drops below a set threshold for some time, it is

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marked as a distraction. This is an indicator that the student is distracted and requires attention.

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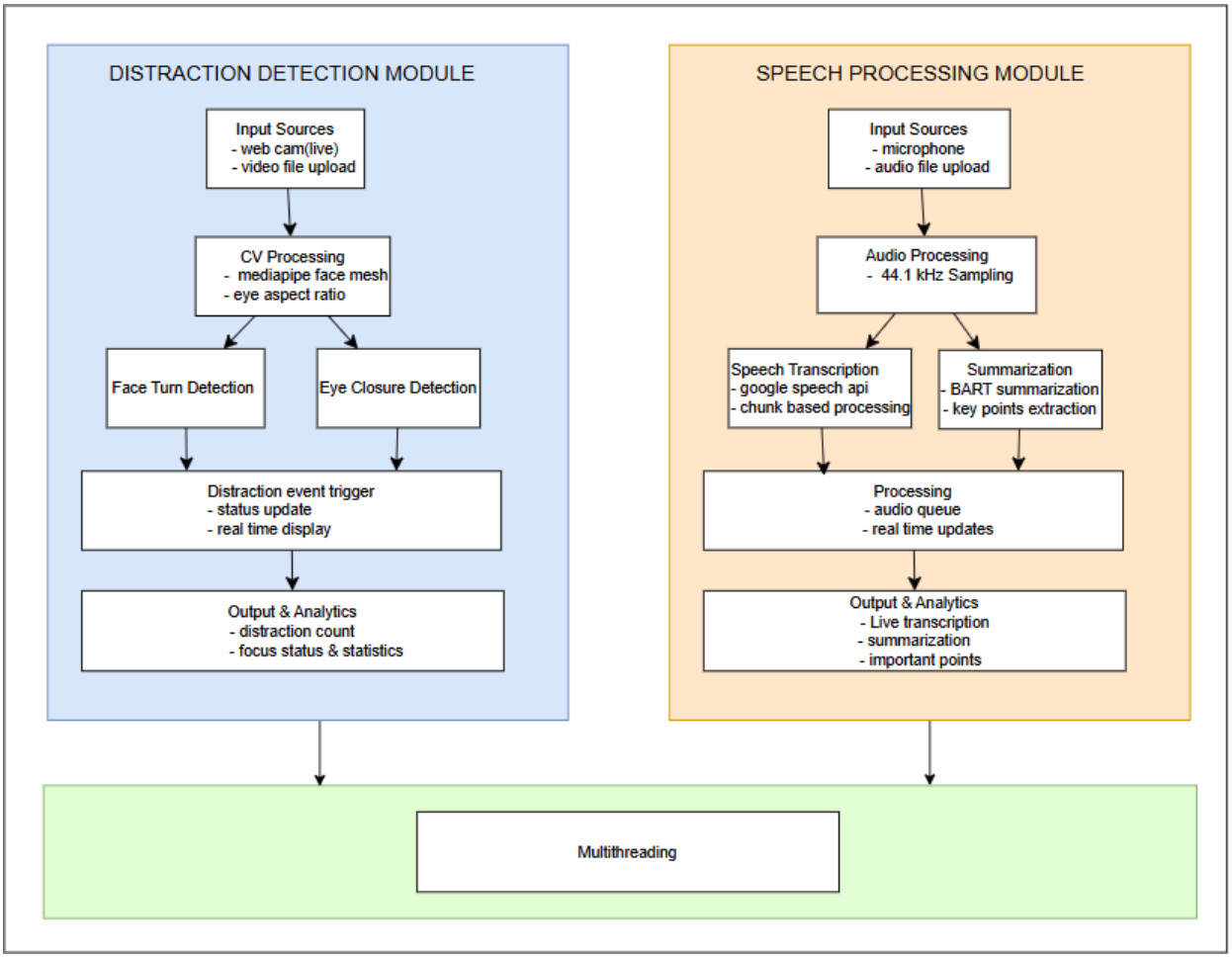


Fig. 1. System architecture showing Distraction Detection and Speech Processing modules. It captures webcam and audio input to detect focus loss and generate real-time transcriptions with summaries.

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Face Orientation Detection In addition to the eye tracking [Jyotsna et al., 2023], the system also tracks the orientation of the face using the position of the student's nose. A deviation from the center of the frame indicates that the student's face has turned away, which is another sign of distraction. This additional measure ensures that the system accurately detects distractions related to both eye movement and face orientation.

3.2 Speech Processing

The *Speech Processing* [Sethiya and Maurya, 2025] module enables the system to transcribe spoken content or live audio or any pre-recorded audio file into text and generate summaries of the transcribed content. This module helps students by providing real-time written notes and important highlights of the lecture useful for their studies.

Audio Capture and Speech Recognition The system captures audio input through a microphone. Using the SpeechRecognition library, the system transcribes the captured speech into text in real- time. The transcription is displayed live on the screen, assisting students in following along with the spoken content.

Audio Segmentation and Transcription For recorded sessions, the system segments the audio into manageable chunks to facilitate easier transcription [D and Fazil, 2024]. The transcription process is performed non-concurrently, ensuring minimal delay in generating the text output.

Text Summarization Once the transcription is complete, the text is passed through the BART model (Bidirectional and Auto-Regressive Transformers) for summarization [Agrawal et al., 2024]. The BART model summarizes the transcribed material, keeping the essential ideas while omitting non-essential information. This gives the student a summarized version of the spoken material, enhancing their attention to key information.

Key Points Extraction In addition to summarizing the text, the system extracts key points from the summarized content. Important sentences and words are highlighted, helping students identify the most critical aspects of the lesson or discussion.

3.3 Parallel Mode

The two components, *Distraction Detection* and *Speech Processing*, are integrated into a coordinated system through multithreading. The *Distraction Detection* module provides real-time feedback to the student by alerting them if they are losing focus, while the *Speech Processing* module ensures that recorded content is transcribed and summarized for easier understanding.In this mode you can get the student's focus percentage during the lecture.

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3.4 User Interface and Accessibility

The system is made with an accessible user interface (streamlit) to accommodate learning-disabled students. The interface gives visual feedback, distraction alerts and status messages. The tran- scribed and summarized text is also presented in an easily accessible manner to facilitate student interaction and can also be downloaded and saved for future access of the notes.

Fig. 2 portrays the User Interface of the system using streamlit for the first module of Distraction Detection and Fig. 3 portrays the User Interface of the system using streamlit for the second module of Speech Processing.

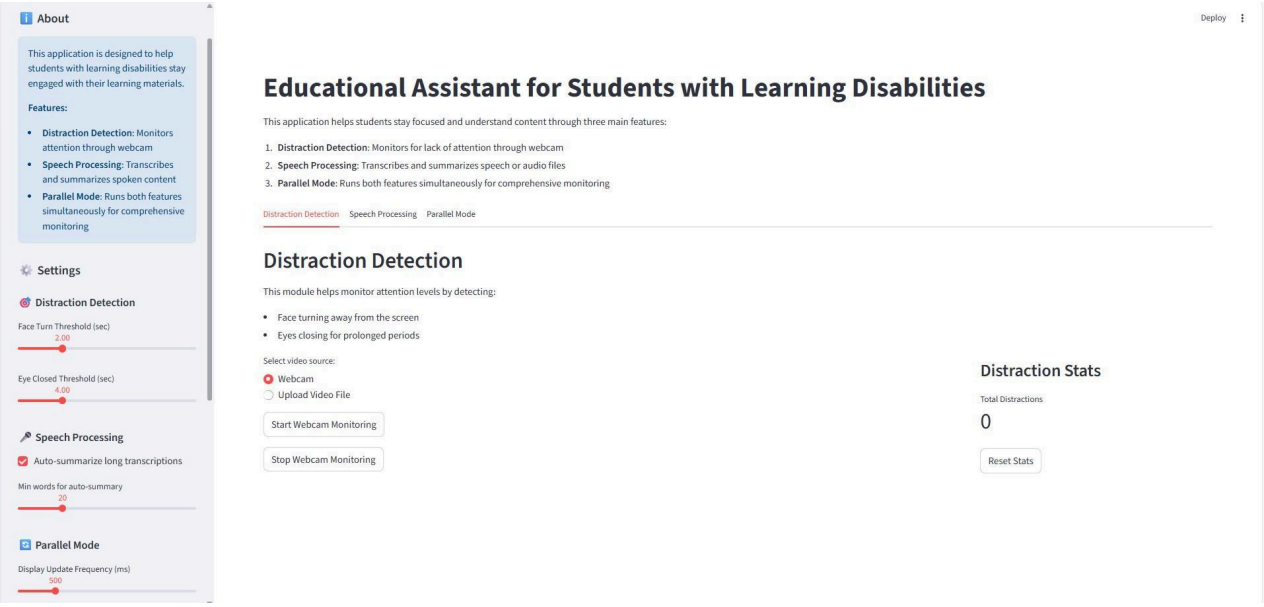


Fig. 2. Streamlit-based interface displaying real-time distraction detection using face orientation and eye aspect ratio analysis.

4 Results

The developed system was seen to accurately perform real-time distraction detection based on face orientation and eye closure analysis through webcam feed and MediaPipe library, accurately in test scenarios. There existed an embedded audio feedback mechanism of timely verbal prompts via pyttsx3 engine to remind students of refocusing as soon as symptoms of distraction were ob- served. The speech-to-text component, which utilized the Google Speech Recognition API, yielded low-latency and accurate transcriptions with high robustness to background noise.

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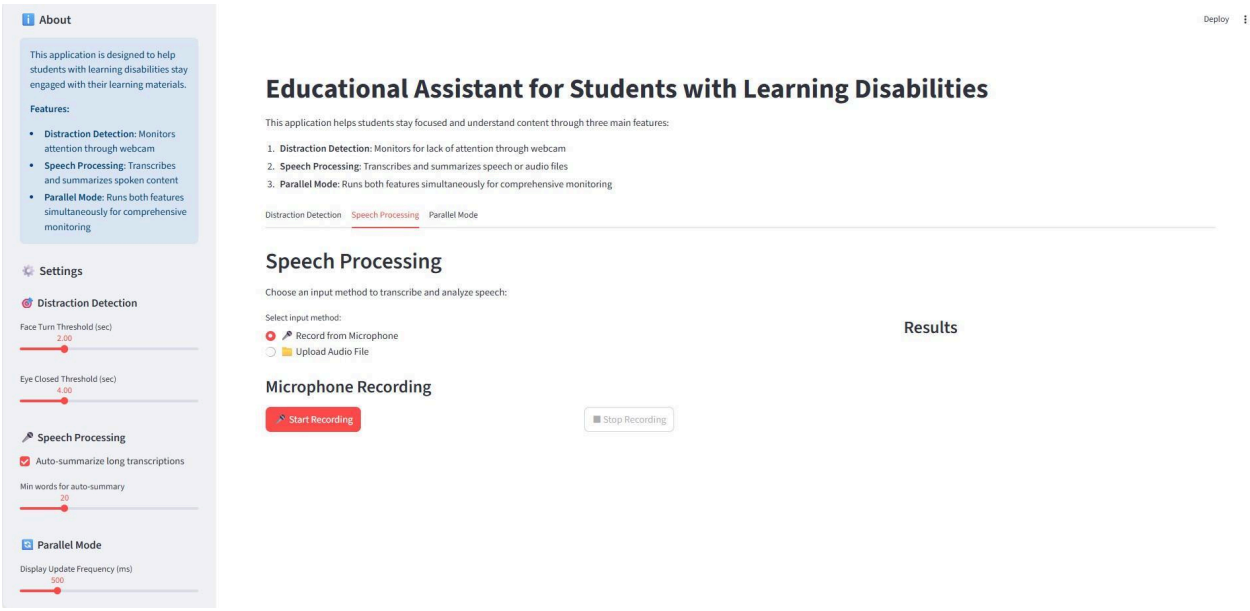


Fig. 3. User interface of the Educational Assistant showing Speech Processing features. It includes audio input options, transcription controls for personalized learning support.

lectures were efficiently summarized using the BART transformer model, producing coherent and concise outputs. Key points were determined by frequency of key words, enhancing content reten- tion.The system also showed smooth multithreaded performance, executing distraction monitoring and speech summarization concurrently with efficient resource management. Last but not least, the implementation guaranteed clean and user-managed shutdown via a shared event flag, helping to make it robust and user-friendly.

The application's Parallel Mode Fig. 4 runs both Distraction Detection and Speech Processing concurrently, enabling full student monitoring. In this mode, the left panel contains the live web- cam feed with distraction detection and the right panel shows the real-time transcription of speech. Results are then consolidated to produce detailed summaries of the user's focus and comprehen- sion, with metrics such as total distractions, words transcribed, session length and overall focus percentage, enabling users to monitor engagement throughout a session.

5 Discussion

Experiments show that the proposed system is effective providing timely assistance to students with learning disabilities as it consists of distraction detection and speech processing. The dis- traction module, utilizing the face mesh and Eye Aspect Ratio (EAR) from MediaPipe was able to detect eyes closure or face orientation signals for lapses in attention, and provide timely alerts through haptic feedback encouraging re-engagement. Google's Speech Recognition API was used by the speech processing module Supporting accurate transcriptions. The speech processing

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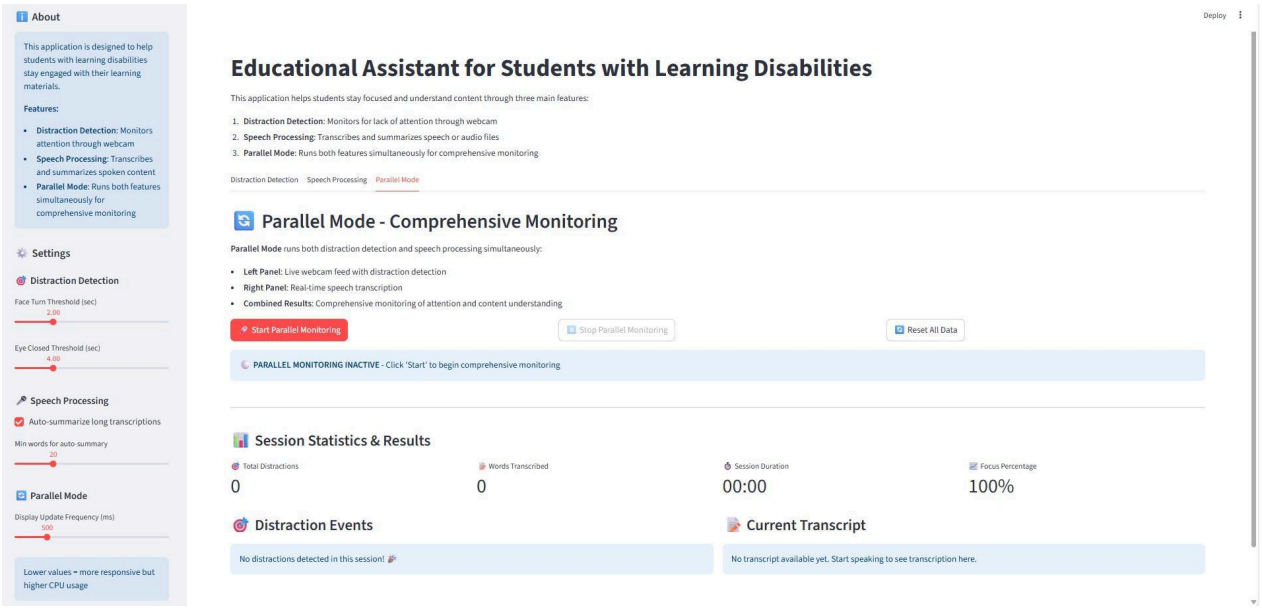


Fig. 4. User interface of the Educational Assistant showing the parallel mode features using multithreading. Distraction Detection and Speech processing module working concurrently.

ule which The results indicated that clear and concise summaries can be written using BART summarization simplifying information retention as well. The system combats existing assistive technologies which normally provide either delayed or passive feedback using Live, actionable interventions. The fact that combined multithreading and a dual-module system architecture prevented any parallelization lag helped. Still, this comes at the cost of sensitivity to background noise for speech input along with consistent lighting conditions for face detection which could impact ro- bustness outside of controlled environments. Even within these limitations, the assistant improves access to learning by tracking attention and improving understanding itself filling an important gap in adaptive education tools.

6 Conclusion

The development of an educational assistant for students with learning disabilities has demonstrated how technology can assist and enhance the learning process of each student who has difficulty with attention and understanding. Through the incorporation of distraction detection using computer vision and speech processing for real-time transcription and summarization, the system provides a solution that caters to major learning engagement elements.According to experimentation,we can assert that the system proves useful in distraction detection and making content more accessible for students by means of automated summaries.Future research will involve fine-tuning these features, introducing customization options, and improving the system's

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plication in online classes. Overall, this study has proven that intelligent systems have much to offer in the way of improving learning achievements among students with learning disabilities.
Future enhancements to our proposed system can include multi languages support to make it more efficient and emotion detection to measure student engagement more effectively. A system that can personalize content based on attention and performance metrics. Multi-device support and data continuity would be made possible by cloud integration, and cross-platform and mobile develop- ment could improve usability on smartphones and tablets. Introducing dashboards for parents and educators will provide a real-time feedback on student engagement. Additionally, ensuring data privacy and ethical compliance and integrating with existing platforms like Google Classroom or zoom would enhance usability and security.

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8 Conflict of Interest

The authors declare that they have no academic or intellectual biases that could have influenced the work reported in this paper.

9 Author Contributions

- Author 1: Writing – original draft, Software, Methodology, Conceptualization.
- Author 2: Writing – review & editing, Validation, Conceptualization.
- Author 3: Writing – review & editing, Validation, Methodology, Conceptualization.
- Author 4: Writing – review & editing, Validation, Supervision, Conceptualization.

10 Plagiarism Report Declaration

The authors attest that they have not used an AI tool to generate any part of the text, the plagiarism is 7% detected from Turnitin, and that all data relevant to this study are included in this article. The authors declare that they have no conflicts of interest regarding the publication of this paper; data analysis and writing were conducted without any academic, financial, or personal conflict affecting the results or interpretations in this study.

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